When Subject Leakage Becomes a Shortcut: Verification, Refusal, and Scoped Repair in EEG Representations

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Abstract

Subject and domain information is strongly decodable from EEG representations, and it is often assumed that decodable leakage means the model relies on a harmful shortcut. We argue this inference is unlicensed: probing shows what is *encoded*, not what is *used*. We define a six-level functional-shortcut-reliance (FSR) ladder — detectability, reducibility, erasability, task coupling, functional reliance, and target consequence — and require any shortcut claim to relate levels rather than leap from a low one to a high one. We instantiate the ladder on real EEG. First, measured leakage magnitude does not certify functional reliance: task-head alignment is positively associated with reliance while recomputable leakage carries the wrong sign. Second, subject signal is erasable, yet no eraser achieves a proven target benefit across 40 held-out-target cells, and aggressive erasure can harm the target. Third, a direct branch-local audit of a fusion backbone shows its spatial branch is the strongest subject-leakage and load-bearing candidate, but removing the subject subspace *hurts* the target — so FSR *refuses* to call it a harmful shortcut; it is task-entangled. Fourth, on injected positive controls we detect, localize, and exactly attribute a known harmful shortcut, and we find a sharp *repair scope boundary*: a controlled first-moment deterministic offset is repairable by a deployable target-unlabeled mean alignment (narrowly, construction-matched), whereas a controlled second-moment stochastic perturbation is not repaired even by an oracle-directed covariance-shrinkage. We characterize this as a *recoverability boundary*: a deployable target-unlabeled operator inverts only corruptions that are batch-deterministic, task-separable, and identifiable, and refuses otherwise. FSR is a shortcut *verification* framework, not a domain-generalization method: it makes explicit what measurement licenses what control conclusion, and it bounds where repair is — and is not — possible.

1 Introduction

EEG representations carry strong subject- and recording-identity information: a linear or shallow probe recovers subject identity with high accuracy, and this signal is often entangled with the cohort/label structure of a dataset [Lin et al., 2026]. A common inference follows: if subject information is decodable from the representation, the model must be *relying* on it as a harmful shortcut, and removing it should improve cross-subject generalization. This paper argues that the inference is unlicensed and provides the audit that separates its premises from its conclusion.

The gap is the same one identified in the probing literature: a probe measures what is *encoded*, not what is *used* [Elazar et al., 2021]. Concept-erasure methods sharpen the point — they can remove a linearly-decodable attribute in closed form [Belrose et al., 2023] or by nullspace/adversarial projection [Ravfogel et al., 2020, 2022] — but removability is a statement about the representation, not a guarantee of a downstream benefit. And the domain- generalization literature warns that a moving invariance proxy does not imply a generalization gain once selection and baselines are controlled [Gulrajani and Lopez-Paz, 2021].

We formalize this as a six-level *functional-shortcut-reliance* (FSR) ladder (Figure 1): detectability (L1), reducibility (L2), erasability (L3), task coupling (L4), functional reliance (L5), and target consequence (L6). A harmful functional shortcut requires domain information that is simultaneously measurable, localizable, interventionable, functionally relied upon, and harmful in target consequence; a claim about a shortcut must therefore be a relationship *between* levels, never a leap from L1 to L5/L6.

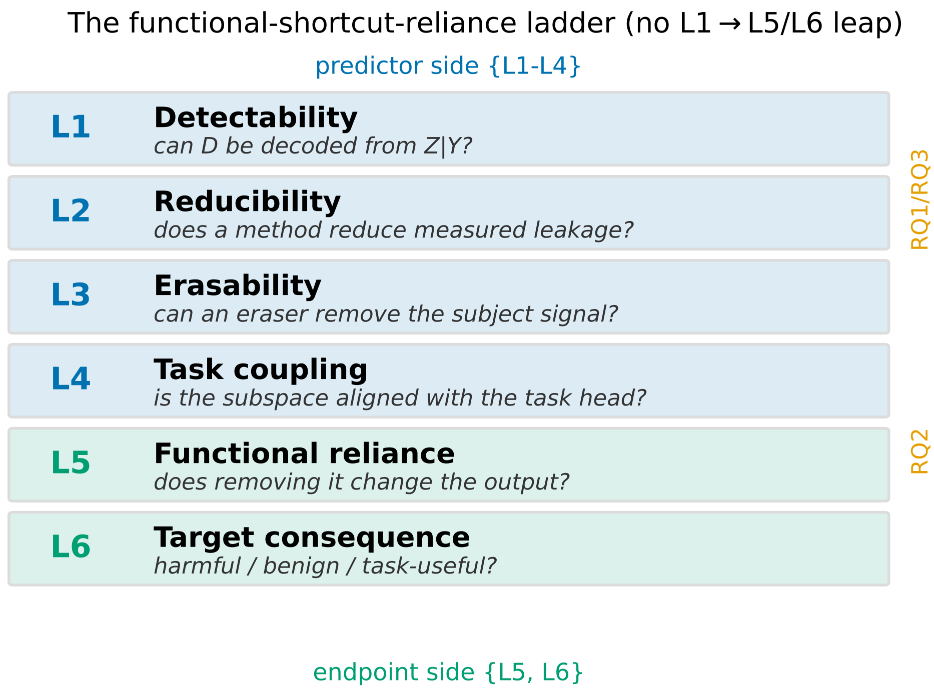

Figure 1: The six-level functional-shortcut-reliance ladder. Predictor-side levels {L1–L4} and endpoint-side levels {L5,L6}. A shortcut claim must relate levels; inferring L5/L6 from L1/L2 is forbidden.

Contributions. Re-analysing frozen artifacts (no new training) across a graph-CMI diagnostic, a frozen-feature concept-erasure study, a branch-fusion backbone re-fit, and injected positive controls, we show: (1) **measurable $\neq$ relied-upon** — measured leakage magnitude does not certify functional reliance; task-head alignment is positively associated with reliance while recomputable leakage carries the wrong sign; (2) **erasable $\neq$ beneficial** — subject signal is erasable, but erasure strength does not certify a target benefit (no eraser achieves a proven gain in 40 held-out-target cells); (3) **natural verification refuses blind repair** — a direct branch-local audit finds the spatial branch is the strongest subject-leakage and load-bearing candidate, yet removing its subject subspace *hurts* the target, so FSR refuses the harmful-shortcut label (it is task-entangled); (4) **repair has a scoped boundary** — on injected controls we detect, localize, and exactly attribute a known harmful shortcut, and find that a controlled first-moment deterministic offset is repairable by a deployable target-unlabeled mean alignment (narrowly, construction-matched) while a controlled second-moment stochastic perturbation is not repaired even by an oracle-directed covariance-shrinkage. FSR is a verification framework, not a domain-generalization method; a companion premise is that conditional-invariance/CMI *control* on these representations is already a closed negative.

2 The FSR audit ladder

Let Z be an EEG representation, Y the task label, and D the domain (subject/recording). Figure 1 defines the six levels.

- **L1 Detectability.** Can D be decoded from Z given Y? A label-conditional domain probe / posterior-KL proxy $I_w(Z; D \mid Y)$ (a plug-in linear-probe surrogate, *not* an unbiased mutual information) with a within-label permutation null.

- **L2 Reducibility.** Does a method reduce the *measured* leakage (paired ΔKL / Δprobe accuracy vs a matched baseline)?

- **L3 Erasability.** Can a linear/non-linear eraser remove the subject signal? Post-eraser residual subject decodability for LEACE/INLP/RLACE/mean-scatter, with a same-rank random-k control.
- **L4 Task coupling.** Is the subspace aligned with the task head? The fraction of linear task-head row-space energy in the top-k label-conditional subject subspace (align_k), and, for multi-branch models, branch-ablation drop plus fusion-gate weight.
- **L5 Functional reliance.** Does removing the subspace change the task head's output? Head-replay task-drop $R3$ after removing the top-k subject subspace (source-only fit), with a random-subspace control and an exact-replay firewall.
- **L6 Target consequence.** Is that reliance harmful, benign, or useful? Held-out-target bAcc/NLL/ECE delta and worst-subject delta, with target labels used only for scoring.

Direction contract. Under the naive "leakage-is-shortcut" hypothesis, corr(leakage, $R3$) and corr(subject-removed, target-b should be *positive*. The FSR questions ask whether they are.

Quantitative-inclusion gate. A route may enter a cross-level quantitative test only if it carries ≥ 1 predictor-side level {L1–L4} *and* ≥ 1 endpoint-side level {L5,L6}; otherwise it is tagged support/boundary/protocol/backgrour and informs interpretation only. Of the audited routes, only a small set is quantitatively includable (Table 4).

3 Setup and provenance discipline

All results are re-analyses of *frozen* artifacts; we train, tune, and select nothing on the analysed data. Three pipelines supply the levels. **CIGL** (RQ1/RQ3): a static graph backbone on BCI-IV-2a (BNCI2014_001) and BNCI2015_001, leave-one-subject-out (LOSO), seeds {0,1,2}, providing per-unit align_k, graph leakage, and $R3$. **TOS** (RQ2): frozen-feature erasure on 2a, Lee2019, Cho2017, and Schirrmeister2017, backbones TSMNet [Kobler et al., 2022] (SPD, z=210) and EEGNet [Lawhern et al., 2018] (z=16), erasers LEACE [Belrose et al., 2023], INLP [Ravfogel et al., 2020], RLACE [Ravfogel et al., 2022], mean-scatter, and random-k; source-only fit, held-out-target deploy. **FBCSP-LGG** (RQ4): a filter-bank CSP [Ang et al., 2012] plus learnable-graph backbone with a three-way gated fusion over graph/temporal/spatial branches; branch ablation and gate weights only.

Claim-strength tiers. Every quantitative statement is tagged RECOMPUTED (recomputed from committed per-unit data), RECOMPUTED_SIGN_ONLY (only a recomputable slice confirms the sign), or FROZEN_NOT_RECOMPUTA (carried from a frozen output whose per-unit inputs are unavailable — support only, never a reproduction). A fail-closed validator enforces the schema, and a reproduction script re-derives the headline and halts if it drifts beyond tolerance.

Target-label firewall. Target labels y_T are used only to score the L6 endpoint; they never enter probe/eraser fitting, model selection, hyper-parameter selection, or early stopping. Each route carries a fit-tag in {NO, YES_FORBIDDEN, AUDIT_ONLY, UNKNOWN}; every quantitative test uses only NO-tagged routes. One legacy route is YES_FORBIDDEN (a retracted, disclosed target-label leak) and is excluded from all tests (Table 4).

Statistics. Spearman correlations with a paired bootstrap CI (rng(0), 2000 resamples, percentile[2.5, 97.5]); the alignment-vs-leakage comparison uses a signed-difference bootstrap. Mechanism regressions use OLS with a z-scored outcome and predictors plus one-hot dataset/seed/method controls. Robustness includes dataset-stratification, per-seed, per-method, leave-one-dataset-out, within-group rank-residualization, and, for erasure, subset sensitivity across eraser families. All analyses are CPU-only over frozen artifacts.

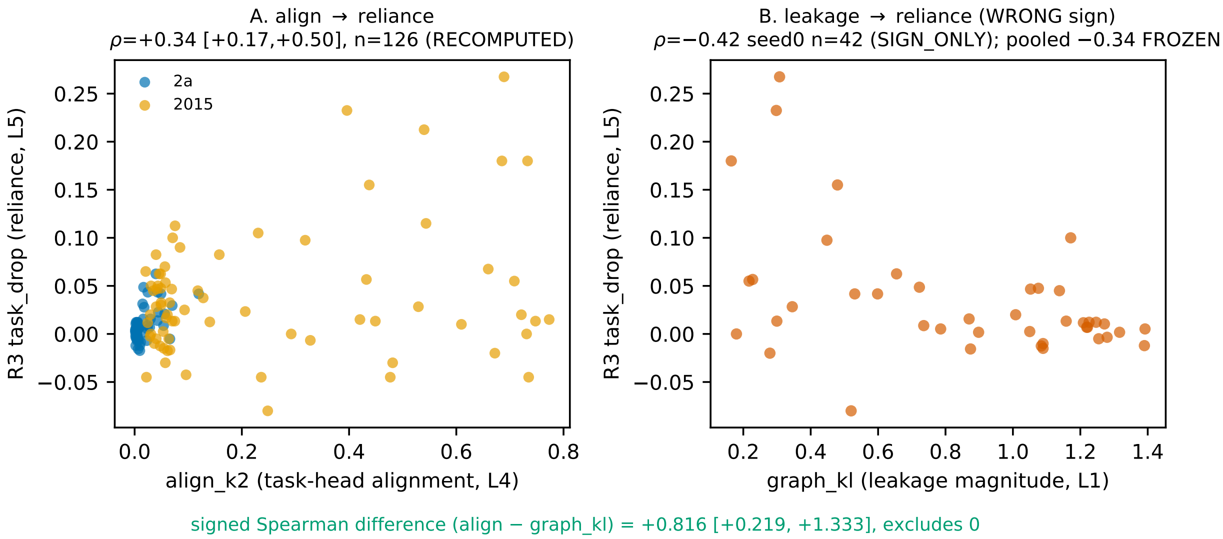

Figure 2: Alignment vs leakage as predictors of functional reliance ($R3$). **A**: task-head alignment is positively associated with reliance (ρ=+0.34 [+0.17, +0.50], n=126, RECOMPUTED); the association is dataset-heterogeneous (2a significant, 2015 not). **B**: recomputable graph leakage carries the wrong (negative) sign (ρ=−0.42 seed0, n=42, RECOMPUTED_SIGN_ONLY); the pooled −0.34 is FROZEN_NOT_RECOMPUTABLE. The signed Spearman difference (align − leakage) = +0.816 [+0.219, +1.333] excludes zero.

4 Measurable is not reliance (RQ1/RQ3)

RQ1 — does measured leakage predict functional reliance? We report three claim-strength tiers. **(RQ1A, RECOMPUTED)** Task-head alignment is positively associated with $R3$: pooled Spearman ρ=+0.34 [+0.17, +0.50], n=126; the association is significant on 2a ($\rho\approx$+0.47) but not on 2015 ($\rho\approx$+0.10), i.e. pooled-positive but not universal. **(RQ1B, RECOMPUTED_SIGN_ONLY)** Graph leakage carries the *wrong* sign: seed0 ρ=−0.42 [−0.68, −0.10], n=42. **(RQ1C, FROZEN_NOT_RECOMPUTABLE)** The pooled leakage correlation −0.34 is carried from a frozen output whose per-fold inputs for two of three seeds were pruned; we do not present it as a reproduction. Thus measured leakage magnitude does not certify reliance, and where recomputable it points the wrong way.

RQ3 — is alignment a better reliance predictor than leakage? The primary test is the signed Spearman *difference* (align − leakage) = +0.816 [+0.219, +1.333], which excludes zero: alignment is more positively associated with reliance than leakage. In a paired standardized regression on the recomputable seed0 units, alignment is correctly signed (β=+0.26) and leakage wrongly signed (β=−0.37); with a dataset control, the alignment partial effect is positive but not individually significant, reflecting the RQ1A heterogeneity. We therefore state that *task-head alignment is closer to functional reliance than raw leakage in this frozen diagnostic* — and explicitly *not* that align$_k$ is a validated reliance estimator.

5 Erasable is not beneficial (RQ2)

Subject signal is erasable: LEACE drives linear subject decodability to chance on both backbones (a nonlinear MLP residual persists), and INLP drives it to chance entirely. The question is whether that removability confers a target benefit.

No certified benefit. Deploying source-fitted erasers to held-out target subjects, we find **no eraser achieves a proven target-bAcc gain in any of 40 cells** (benefit_claimable = 0/40 under a rule requiring a positive point estimate whose 95% CI lower bound exceeds zero, no task-collapse, no binary-latent harm,

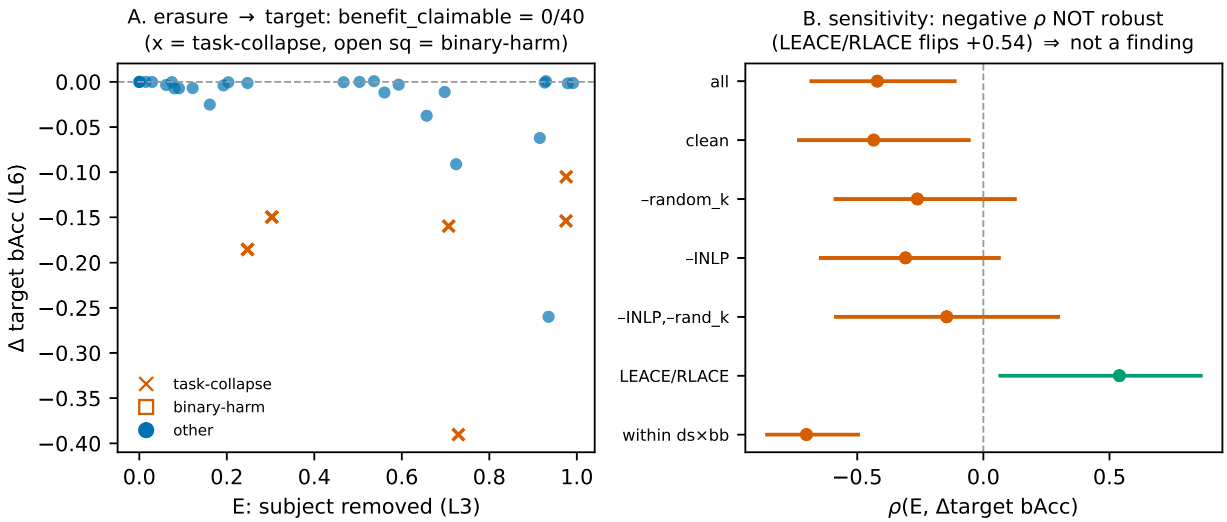

Figure 3: Erasure strength vs target benefit across 40 held-out-target cells. **A**: no eraser achieves a proven target-bAcc gain (benefit_claimable = 0/40); many cells task-collapse (INLP) or harm a binary-latent target (LEACE/RLACE on Lee/Cho EEGNet). **B**: the apparent negative ρ(subject-removed, Δtarget bAcc) is *not robust* — it flips to +0.54 on the principled-eraser subset (LEACE/RLACE) and is non-significant when INLP or random-k are dropped; it is therefore not reported as a finding.

and no random-k-matched non-specific NLL move; Figure 3A). Erasure can actively harm the target: INLP over-erases and drives task to chance in several cells, and on the compact binary EEGNet latent (Lee2019, Cho2017) LEACE and RLACE drive task-bAcc to chance, a task-safety failure. On 2a-TSMNet the small NLL improvement under LEACE is matched by a same-rank random removal that leaves the subject fully decodable, so that NLL movement is non-specific regularization, not domain removal.

The negative correlation is not a finding. Pooled across all 40 cells, ρ(subject-removed, Δtarget bAcc) is negative and excludes zero. A sensitivity battery (Figure 3B) shows this is not robust: it is non-significant when INLP or random-k are removed, and it *flips to* +0.54 (excluding zero) on the principled-eraser subset (LEACE/RLACE only). The negative sign is driven by INLP over-erasure and the random-k anchor, not by a "more removal, worse target" law. We therefore report only the robust result — the absence of a certified benefit — and do not headline a negative association. The upshot: subject signal is erasable, but erasure strength does not certify target benefit.

6 Natural branch-local verification refuses blind repair (RQ4)

A multi-branch backbone lets us ask whether leakage *means* something different in a load-bearing branch than elsewhere — and, decisively, whether the strongest natural leakage candidate is actually a harmful shortcut. Unlike the frozen-artifact stage, we now measure the full branch-local ladder directly: a minimal ERM re-fit of the same FBCSP-LGG recipe (no CMI, no architecture or hyperparameter search, no target-label fit) reproduces the frozen deployment metrics (target bAcc 0.358/0.599 vs. 0.349/0.608, $|\Delta|$=0.009) and, additionally, dumps per-branch latents and best-epoch checkpoints, so L1–L6 are computed per branch on real EEG. The target-label firewall is clean on all 21 folds (labels enter only final scoring; branch selection, probes, and subspace fits are source-only).

The spatial branch is the strongest candidate on every low level, yet not harmful. The spatial branch is the most subject-decodable (L1 0.92/0.82), the most load-bearing (zeroing it drops task-bAcc by +0.086 pooled; highest fusion-gate weight), and functionally coupled (L5: erasing its source-estimated

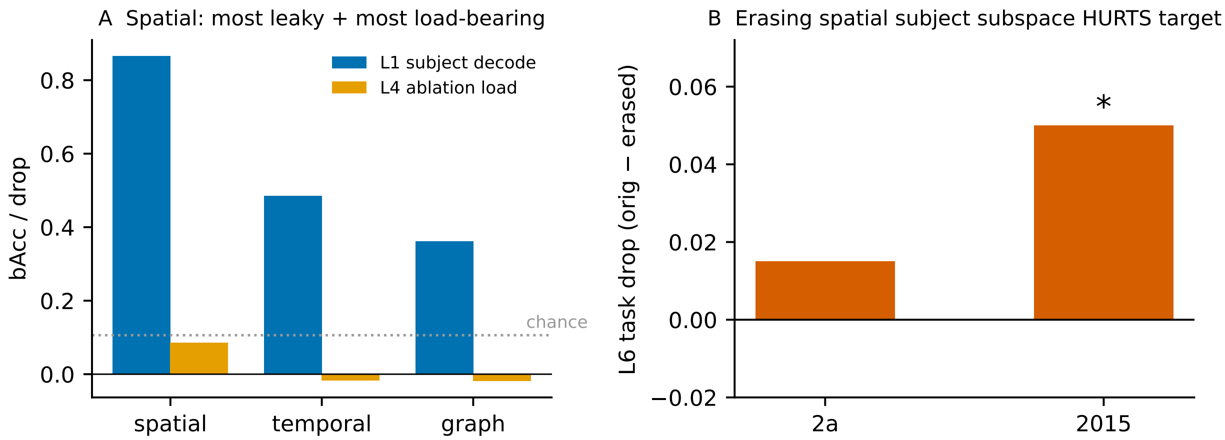

Figure 4: Direct branch-local audit of the FBCSP-LGG fusion backbone (Phase 4B refit, 21 leave-one-subject-out folds on BNCI2014_001/2015_001). The spatial branch is simultaneously the most subject-leaky (L1 subject decodability 0.92/0.82), the most load-bearing (L4 ablation drop +0.086; highest fusion-gate weight), and functionally coupled (L5 head-replay specificity). Yet *removing* its subject subspace *hurts* the target (L6 task drop positive, +0.050* on 2015): the erasing-helps criterion fails. Verdict: NO VERIFIED HARMFUL BRANCH SHORTCUT.

subject subspace changes the head's output specifically, beyond a random-subspace control). By L1–L5 alone it looks like the canonical harmful shortcut. But the target consequence inverts the inference: removing the subject subspace *raises* the erased-model's task drop rather than lowering it (L6 task_drop= + 0.015 on 2a, +0.050* on 2015), i.e. *erasing the subject signal hurts the target.* The subject subspace is task-entangled — a task-useful reliance, not a harmful one.

Verification, not a leap. FSR therefore **refuses** to call the strongest natural subject-leakage branch a harmful shortcut: the verdict is NO VERIFIED HARMFUL BRANCH SHORTCUT (C16). This is the ladder working as designed — a claim that stops at L1/L5 ("subject signal is strongly decodable and functionally coupled, so remove it") would be wrong here, because the L6 test shows the removal is harmful. We can state that the spatial branch is load-bearing (C6) and that its subject signal is measurable and coupled, but we *cannot* say "spatial leakage is harmful" or "erase spatial subject directions to improve DG." On natural EEG, the honest conclusion is a refusal.

7 Repair has a scoped boundary (controlled positive controls)

The natural result is a refusal; to test the *machinery* of detection, attribution, and repair we turn to controlled positive controls — known harmful shortcuts injected into the frozen spatial branch. Everything here is target-unlabeled and source-fit: target labels enter only the final score, and injection tokens, strengths, and repair hyperparameters are chosen from source data alone.

Detect, localize, attribute. An injected subject-token shortcut (a per-subject additive direction with a class-directed component) is detected as induced target harm (+0.041/ + 0.066 bAcc at strengths α=1/2), correctly *localized* to the injected spatial branch (harm 0.041 vs. 0.021 graph, 0.020 temporal), and *exactly attributed* (subtracting the known per-sample token recovers the harm fully, recovery 1.0). The verification–localization–attribution chain works on real EEG representations (C11).

Erasure is not the repair; first-moment alignment is, within a narrow scope. Erasing the injected subspace — even an oracle that knows it — makes the target *worse*, because the harmful direction is task-coupled (the same task-entanglement seen naturally). A counterfactual-consistency adapter ties a random

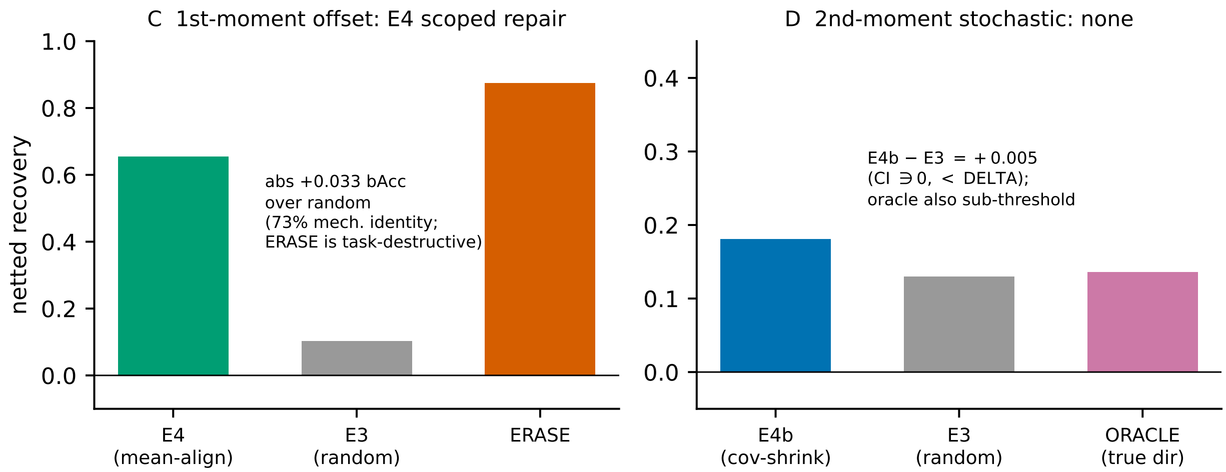

Figure 5: Controlled-repair scope. A known harmful shortcut is injected into the spatial branch and repair is attempted with target-unlabeled, source-fit primitives (target labels score only). **Left:** a *first-moment* constant-offset injection is repaired by mean alignment (E4), netting a shortcut-specific +0.033 bAcc over a random control — but 73% of the "recovery" is a mechanical identity (a first-moment aligner inverts a first-moment offset by construction), and it fails leave-one-dataset-out. **Right:** a *mean-null second-moment* (stochastic) injection is *not* repaired by covariance-shrinkage (E4b) at the source-selected operating point — E4b does not beat a random-direction control, and even an oracle that shrinks the true injected direction is sub-threshold.

control. The mechanistically-matched primitive is different: because the injected token is a *constant per-batch offset*, a deployable *first-moment mean alignment* (E4, align the target feature mean to the source, target-X-only) reverses it. Under a corrected, pre-registered, clustered-bootstrap protocol on eight fresh injection seeds, E4 yields a shortcut-specific netted gain of +0.033 bAcc (clustered CI $[0.009, 0.058]$), beating a random-direction control (C12). The claim is deliberately narrow: 73% of the pooled recovery is a *mechanical identity* (a first-moment aligner inverts a first-moment offset by algebra; the non-identity recovery is 0.68), the effect is carried by one of the two datasets and *fails leave-one-dataset-out*, and E4 overshoots the clean target — it is a generic domain-mean alignment, not a surgical shortcut remover. We report it as STRONG WITHIN A CONTROLLED FIRST-MOMENT SCOPE, never as general repair.

Second-moment shortcuts are not repaired at the operating point. We then inject a strictly *mean-null second-moment* shortcut (per-sample zero-mean variance along a class-directed direction; harm +0.031 bAcc, mean displacement exactly 0). Here first-moment alignment is powerless by construction, and the matched primitive is covariance-shrinkage (E4b, shrink the target's excess-variance direction toward the source level). E4b does *not* beat a random-direction shrinkage control (E4b − E3 = +0.005 bAcc, clustered CI $[-0.005, 0.014]$, below the 0.02 bar), and — decisively — an *oracle* that shrinks the *true* injected direction is also sub-threshold (+0.004, CI $[-0.005, 0.012]$), while the estimator does recover the true direction (overlap 0.71). The failure is therefore a genuine weakness of covariance-shrinkage, not mis-estimation: a stochastic per-sample perturbation that has already scattered samples across the decision boundary is not recoverable from batch-level second-moment statistics. A direction-specific advantage clearing the bar appears *only* in an injection-dominant regime that the source-only strength rule excludes; at the honest operating point the verdict is NONE (C17, C14: not established).

The scope boundary. Table 1 summarizes the boundary. A controlled *first-moment deterministic* offset is repairable (narrowly, construction-matched); a controlled *second-moment stochastic* perturbation is not (even oracle-directed); and the *natural* branch-local subject signal is refused as task-entangled (§6). The deployable repair family is first-moment specific; repair of anything beyond a deterministic offset — and of natural or learned reliance — remains open (§10).

Shortcut regime	Repair primitive	Outcome	Verdict
Natural branch-local (spatial)	(verification)	subspace removal *hurts* target	REFUSE (task-entangled)
Injected 1st-moment (offset)	mean alignment (E4)	netted +0.033 bAcc, beats random	SCOPED-STRONG†
Injected 2nd-moment (variance)	covariance-shrink (E4b)	$\leq$ random; oracle sub-DELTA	NONE
Learned reliance (PC2)	—	GPU paused; not eligible	FUTURE WORK

† construction-matched: 73% mechanical identity; fails leave-one-dataset-out; deployable, target-X-only.

Table 1: Repair scope matrix. Deployable target-X-only repair is confined to first-moment deterministic offsets; second-moment stochastic perturbations and natural/learned reliance are not repaired.

8 Recoverability: what determines whether a shortcut can be repaired

Results 6–7 are not three isolated outcomes; together they characterize *when* a verified harmful shortcut is repairable by a *deployable, target-unlabeled* operator. We make the boundary explicit. The deployable repairs we use are **target-X-only batch-statistic affine operators** $R(z) = Az + b$ that apply the *same* map to every sample, with A, b measurable from the target batch's moments $\{\hat{\mu}_T, \hat{\Sigma}_T\}$ and source references (E4 is the translation case, E4b the linear case). Within this class, a clean boundary holds.

> *Characterization (within the batch-affine operator and affine-in-z corruption classes).* A corruption $C : z_i \mapsto z_i + p_i$ is invertible by a target-X-only batch-affine operator iff C is an affine function of z — a batch-constant offset (undone by a translation) or a fixed *full-rank* linear map (undone by a matrix). An additive per-sample perturbation *independent* of z is not invertible, because the per-sample realization is not present in the batch statistics.

This places the empirical results into non-exclusive *obstruction classes* (Table 2). A **first-moment deterministic** offset is affine-in-z and, when task-separable and identifiable, is recoverable (**R0**; E4, Result 7); the recovery is narrow precisely because the offset is conflated with the genuine domain-mean shift. A **per-sample stochastic** perturbation independent of z is *not* invertible by any batch-affine operator, even with the true direction (**R1**; E4b fails even oracle-directed, Result 7) — and this is a genuine non-invertibility, not mis-estimation, since the estimator recovers the direction yet the operator cannot undo the per-sample noise. A **task-coupled** nuisance subspace makes erasure destroy task signal (**R2**; the natural spatial branch, Result 6, and the injected oracle-subspace erasure both fail). A shortcut that is **under-identified from target-X** — a learned reliance or a concept shift — cannot be identified, let alone repaired, from the marginal alone (**R3**).

Head-only learned-reliance bridge. To bridge the controlled injected shortcuts above and a full learned-reliance stress test, we ran two CPU-only head-level probes on the frozen Phase-4B EEG latents (retraining *only* a linear head; target labels used for final scoring only). Source subject-class *prevalence reweighting* on true labels induced the subject–class correlation exactly, but did *not* induce a learned subject reliance: the task signal remained a sufficient statistic and the learnability gate failed closed. A stronger subject-keyed, *task-conflicting* relabeling (global label histogram held exact) was *learnable in-sample* — the head fit the relabeled subset far beyond a task-only floor (0.70 vs 0.20) — yet its held-out and target true-task harm did *not* exceed a subject-scrambled control and was *not* localized to the subject subspace (it beat only a matched random-noise control, on both datasets). Thus, in this head-only setting, neither prevalence skew nor subject-keyed task-conflict turned naturally present subject leakage into transferable, subject-structure-specific harmful reliance; both sit in the under-identified region (**R3**), and neither licenses the paused learned-reliance refit. These probes do *not* prove that learned subject shortcuts cannot occur; they show that under the cheapest controlled head-level manipulations, the available natural subject signal is not readily weaponized without stronger information or a different mechanism.

Two consequences frame the paper's contribution. First, repairability is a property of the *corruption structure* $\times$ *operator class* $\times$ *available information*, not of the shortcut's mere detectability: the same subject signal is task-useful naturally (R2, refuse), repairable as a deterministic injected offset (R0), and un-repairable

Class	Obstruction	Invertible by target-X batch-affine op?	Licensed action
R0	none (affine-in-z, task-sep., identified)	yes (E4; narrow, construction-matched)	REPAIR
R1	per-sample noise indep. of z (2nd-moment)	no (even oracle direction)	REFUSE / NEED PER-SAMPLE INFO
R2	task-coupled subspace ($S \not\perp T$)	no (erasure destroys task)	REFUSE / TASK-PRESERVING INVERSE
R3	under-identified from target-X (learned/concept)	not even identifiable	REQUIRE STRONGER CONTRACT
Head-only learned-reliance probes (R3 region; frozen 4B latents, linear head):			
7B prevalence	subject–class correlation, reliance not induced	n/a (task is a sufficient statistic)	NO REPAIR; PC2 UNLICENSED
7C task-conflict	learnable in-sample, harm not subject-structured	n/a (ties the subject-scramble control)	CONTROLLED NEGATIVE

Non-exclusive; the binding (most-restrictive) obstruction sets the action. R0 is the unique cell where none hold.

Table 2: Recoverability obstruction classes (non-exclusive; the binding obstruction sets the licensed action). Deployable target-X-only batch-affine repair reaches only R0; R1/R2/R3 are detected, localized, and refused, and require stronger information (per-sample side information, a task-preserving inverse, or labels/paired/randomized data). "Not repairable" is scoped to batch-affine operators, not proven for all operators.

as a stochastic injected perturbation (R1). Second, the open cases (R1/R2/R3) are exactly where FSR *refuses* or requires a stronger information contract — the same discipline (available information → what is identifiable → what action is licensed) that recurs across leakage measurement, erasure, and adaptation. Repair is not a switch to flip; it is an action licensed only inside a recoverability class.

9 Related work

Probing measures encoding, not use. A probe establishes that an attribute is *decodable* from a representation, not that a model *uses* it; amnesic and counterfactual probing make the distinction operational by intervening on the representation and observing behavior, and show that decodability and causal use can dissociate [Elazar et al., 2021]. Our L1↔L5 separation is the EEG realization of this: a subject probe measures detectability, while head-replay $R3$ measures whether removing the subspace actually changes the task head's output.

Concept erasure: removed is not beneficial. A family of methods removes a linearly-decodable concept from a representation: LEACE erases it in closed form with provably minimal collateral damage [Belrose et al., 2023]; INLP removes it by iterative nullspace projection [Ravfogel et al., 2020]; RLACE by a relaxed adversarial minimax [Ravfogel et al., 2022]; and scatter/moment methods target invariant subspaces through a generalized eigenproblem [Ghifary et al., 2017]. These are precisely our L3 operators. Prior work establishes *removability*; we deploy the same operators source-to-target and test the downstream consequence, finding that erasure strength does not certify a target benefit.

Domain generalization needs selection rigor. Under careful model selection with a strong ERM baseline, many invariance-based methods fail to beat ERM, and apparent gains often reflect selection rather than a robust effect [Gulrajani and Lopez-Paz, 2021]. This is the direct warning against reading a moving invariance proxy as a generalization gain, and it motivates our refusal to propose a new method: we audit whether the measured proxies even predict the reliance/consequence they are assumed to control.

EEG subject identity is strong and entangled. Audits of EEG foundation models report that subject identity is highly decodable and can confound label/cohort structure, an "identity trap" [Lin et al., 2026].

Because the signal is strong, equating "decodable subject" with "harmful shortcut" is tempting; our audit is designed to separate the two.

EEG decoders and branch structure. The backbones we analyse are standard: filter-bank common spatial patterns for oscillatory motor-imagery features [Ang et al., 2012], the compact convolutional EEG-Net [Lawhern et al., 2018], and the SPD/tangent-space TSMNet with domain-specific normalization [Kobler et al., 2022]. Branch-locality (a gated fusion over graph/temporal/spatial branches) lets us ask whether leakage means something different in a load-bearing branch: a direct branch-local audit (§6) finds the load-bearing branch is the most subject-leaky, yet removing its subject subspace hurts the target — so measurable, load-bearing leakage need not be a harmful shortcut.

Decompose a joint effect into pathways. A single unlabeled adaptation number conflates a geometry pathway and a prevalence/decision-prior pathway and admits an exact decomposition [Anonymous, 2026b]. FSR applies the same measurement-then-decompose discipline to a different object — leakage, erasure, task-coupling, functional reliance, and target consequence — rather than collapsing them into a single "shortcut" scalar.

10 Limitations

We state the limitations plainly; for an audit paper they are part of the contribution.

- **Provenance limit.** The pooled leakage→reliance correlation is not recomputable (per-fold leakage for two of three seeds was pruned); it is reproduced only at seed0 (sign) and carried as FROZEN_NOT_RECOMPUTABLE at pooled. Only alignment→reliance is fully recomputed.
- **Heterogeneity / small n.** The alignment→reliance association is significant on 2a but not on 2015; the alignment-vs-leakage contrast is at seed0, n=42, with non-significant within-group partial betas.
- align_k **is not a validated estimator**, only closer to reliance than raw leakage.
- **The RQ2 negative correlation is not a finding** — it is a non-robust confound of over-erasure and the random-k anchor (Figure 3B); the RQ2 result is the absence of a certified benefit (0/40).
- **The natural-verification refit is minimal ERM**, reproducing the frozen deployment metrics ($|\Delta|$=0.009); it is an instrument to measure the branch-local ladder, not a tuned model, and its conclusion (C16) is a refusal, not a repair.
- **The first-moment repair is construction-matched and narrow** (C12): 73% of the pooled recovery is a mechanical identity (a first-moment aligner inverts a first-moment offset), the effect is carried by one dataset and fails leave-one-dataset-out, and E4 overshoots the clean target — we report it as STRONG WITHIN A CONTROLLED FIRST-MOMENT SCOPE and lead with the absolute +0.033 bAcc gain, not the ratio.
- **Second-moment repair is unresolved** (C14, C17): at the source-selected operating point covariance-shrinkage does not beat a random-direction control, even oracle-directed; a direction-specific advantage appears only in an injection-dominant regime we exclude. We do not claim second-moment shortcuts are unconditionally unrepairable.
- **Injected, not natural, controls.** The repair results use *injected* positive controls to test the machinery; they do not speak to natural or learned reliance, and must not be conflated with the natural refusal (§6).
- **Head-only probes are mechanism tests, not natural-label-noise claims.** The two head-only learned-reliance probes (§8) use controlled source-label manipulations on frozen latents; they do *not* prove that learned subject shortcuts cannot occur, and must not be read as a natural-label-noise or "resists weaponization" claim. They show only that two cheap controlled mechanisms (prevalence skew, subject-keyed task-conflict) do not license the paused PC2 refit or a natural weaponization claim.

- **Two datasets.** The binding robustness axis is leave-one-dataset-out; with two preset-ready datasets it is under-powered (and 4F fails it). A *full-backbone* learned-reliance refit (PC2) and a $\geq$ 3-dataset extension are future work, not current results; the head-only probes above are the cheap mechanism substitute, and current evidence does not authorize the GPU refit.
- **Proxies, not CMI.** L1/L2 use a label-conditional linear-probe posterior-KL surrogate and a linear-probe advantage, not an unbiased mutual information; we say "extractable conditional domain information," not "CMI."
- **Erasability $\neq$ full removal** (LEACE leaves a nonlinear residual); the one target-unlabeled positive we cite (a non-CMI adaptation control) is seed0-only support, not an FSR result.
- **Scope.** Results are on specific backbones and two datasets; generality beyond them is not claimed.

11 Conclusion

We have argued, and shown on real EEG, that a decodable subject/domain signal is not by itself evidence of a harmful functional shortcut. Measurable leakage magnitude does not certify functional reliance — task-head alignment is closer to reliance, though not a validated estimator. Subject signal is erasable, but erasure strength does not certify a target benefit: no eraser achieves a proven gain across 40 held-out-target cells, and aggressive erasure can harm the target. A direct branch-local audit shows the spatial branch is the strongest subject-leakage and load-bearing candidate, yet removing its subject subspace *hurts* the target — so FSR refuses to call it a harmful shortcut; it is task-entangled. And on injected positive controls, repair has a sharp scope: a controlled first-moment deterministic offset is repairable by a deployable target-unlabeled mean alignment (narrowly, construction-matched), whereas a controlled second-moment stochastic perturbation is not repaired even by an oracle-directed covariance-shrinkage, and the natural signal is refused outright.

Measurable is not reliance. Erasable is not beneficial. Natural leakage can be task-entangled, so FSR refuses blind repair; and where repair holds, it holds only within a first-moment deterministic scope.

The contribution is a verification framework, not a new domain-generalization method: the FSR ladder makes explicit what measurement licenses what control conclusion, and it draws the boundary of where repair is — and is not — possible. FSR does not merely say "do not repair"; it says *when* to refuse a blind repair (natural spatial leakage is task-entangled) and *when* a repair holds only within a mechanistic scope (first-moment deterministic offsets, not second-moment stochastic perturbations, not natural or learned reliance). The claim ledger (Table 3) and the route-inclusion ledger (Appendix B, Table 4) record every claim's status, so the boundary between what the evidence supports and what it does not is auditable.

The final picture is not that EEG subject leakage is harmless. It is that a shortcut claim requires an information contract: leakage must be shown to be task-coupled, target-harmful, transferable, and recoverable under the available observations. In our natural and controlled EEG audits — and in the two head-only learned-reliance probes — those conditions *split* rather than collapse into a single leakage score.

A Provenance and reproducibility

All results re-analyse frozen artifacts from three internal repository lines; no model is trained, tuned, or selected on the analysed data. The CIGL graph-CMI line and the TOS erasure line are the internal evidence sources [Anonymous, 2026a,c]; the FBCSP-LGG branch line supplies the branch-load descriptives.

Artifact sources.

RQ	Pipeline	Frozen source (branch @ short-SHA)
RQ1/RQ3	CIGL graph-CMI (2a, 2015; LOSO; seeds 0,1,2)	cigl-r123-scaffold @ d3593c8
RQ2	TOS frozen-feature erasure (2a, Lee, Cho, HGD)	tos @ 1c65d79
RQ4	FBCSP-LGG branch ablation + gate	fbcsp-lgg-spatial-cmi-fusion @ 39c245a

ID	Status	Claim
C1	READY_W/CAVEAT	Measured leakage magnitude does not certify reliance.
C2	READY_W/CAVEAT	Task-head alignment is closer to reliance than raw leakage.
C3	READY	Subject signal is erasable.
C4	READY	Erasure strength does not certify target benefit (0/40).
C5	READY_W/CAVEAT	Random-k falsifies non-specific NLL movement.
C6	READY	Spatial branch is load-bearing.
C7	SUPERSEDED	Branch-local ladder was unmeasured in frozen artifacts (now measured, 4B).
C8	READY	CMI-control remains a closed negative.
C9	SUPPORT	TTA-Control is positive but non-CMI.
C10	READY	FSR is a verification framework, not a new DG method.
C11	READY	FSR detects, localizes, and attributes controlled injected shortcuts.
C12	READY_W/CAVEAT	E4 repairs a controlled first-moment constant-offset shortcut (narrow scope).
C13	FORBIDDEN	E4 as a general / natural shortcut repair.
C14	NOT_ESTAB.	E4b as a controlled second-moment shortcut repair (4G = none).
C15	NOT_READY	PC2 learned-reliance repair (GPU paused; future work).
C16	READY	Natural branch-local subject leakage is not automatically harmful.
C17	READY_W/CAVEAT	Repair scope is first-moment-specific.
C18	READY_W/CAVEAT	Prevalence reweighting does not demonstrate head-level weaponization (7B).
C19	READY_W/CAVEAT	Subject-keyed task-conflict is learnable in-sample but not transferable structure-specific harm (7C).

Table 3: Claim ledger. Statuses are derived from the frozen result artifacts, not hand-set.

Claim-strength tiers. Every quantitative statement carries one tier: RECOMPUTED (recomputed from committed per-unit data), RECOMPUTED_SIGN_ONLY (only a recomputable slice confirms the sign; e.g. per-fold graph leakage survives at seed0 only), FROZEN_NOT_RECOMPUTABLE (carried from a frozen output whose per-unit inputs were pruned — support only, never a reproduction). A fail-closed schema validator and a reproduction script (which halts if the headline drifts beyond tolerance) enforce these.

Target-label firewall. Target labels score the L6 endpoint only; they never enter probe/eraser fitting, subspace fitting, model selection, hyper-parameter selection, or early stopping.

Route family	fit uses target y?	eval uses target y?
CIGL (RQ1/RQ3)	NO (source-only)	YES (scoring)
TOS erasers (RQ2)	NO (source-only)	YES (deploy scoring)

Every route in the audit carries a fit-tag in {NO, YES_FORBIDDEN, AUDIT_ONLY, UNKNOWN}; only NO-tagged routes enter quantitative tests, and the single YES_FORBIDDEN route (a retracted, disclosed target-label leak in a legacy line) is excluded from all tests (Appendix B).

RQ4 checkpoint search (provenance of the blocked status). Answering RQ4 quantitatively requires per-branch leakage (L1) and per-branch reliance (L5), neither of which is on disk; more fundamentally, *no trained checkpoint exists*. A read-only search of the compute store (`/projects/.../yinghao`, maxdepth

6), the user home, and the git branches returned zero matching FBCSP-LGG checkpoints, consistent with a training pipeline that persists only scalar summaries and holds the best-epoch state in memory without serializing it. RQ4 is therefore reported as blocked; producing the metric would require retraining, which we do not do.

B Route inclusion ledger

The audit spans 37 routes across the repository's measurement/control lines. Under the Phase-1 quantitative-inclusion gate (Section 2), a route enters a cross-level test only if it carries ≥ 1 predictor-side level {L1–L4} *and* ≥ 1 endpoint-side level {L5,L6}. Six routes qualify; the remainder are support/boundary/protocol/background and inform interpretation only. No target-label-fit route enters any quantitative test.

Route	Predictor	Endpoint	Inclusion
CIGL	L1—L2—L4	L5—L6	INCLUDED
TOS_mean_scatter	L1—L3—L4	L6	INCLUDED
TOS_LEACE	L1—L3—L4	L6	INCLUDED
TOS_INLP	L1—L3—L4	L6	INCLUDED
TOS_RLACE	L1—L3—L4	L6	INCLUDED
TOS_random_k	L3	L6	INCLUDED
31 other routes: SUPPORT/BOUNDARY/PROTOCOL/BACKGROUND_ONLY (no target-label fit enters any RQ)			

Table 4: Route inclusion. Only routes with a predictor+endpoint pairing enter quantitative tests; no target-label-fit route does.

The excluded routes fall into four tags: SUPPORT_ONLY (a predictor or endpoint level is present but not the pairing a specific RQ needs — e.g. FBCSP branch load is L4-only), BOUNDARY_ONLY (a boundary/decision result such as the erasure refusal gate), PROTOCOL_ONLY (an executed protocol without an efficacy endpoint), and BACKGROUND_ONLY (context lines, e.g. the prior-decoupled adaptation decomposition). The full per-route ladder mapping, missing-metric policy, and target-label tags are released with the artifacts.

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